

ANIMESH DHIMAN

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PROFESSIONAL SUMMARY

Computer Science undergraduate at Thapar Institute (2027) who ships production AI systems end to end from a deep learning paper under review at **Elsevier** to three live, deployed products: an on-device AI assistant, an explainable resume ranking engine, and a government scale identity resolution platform. Interested in AI Safety: mechanistic interpretability, adversarial robustness, and LLM evaluation.

EDUCATION

Thapar Institute of Engineering and Technology

2023 – 2027

Bachelor of Engineering in Computer Science

Patiala, Punjab

TECHNICAL SKILLS

Programming	Python, C++, C, SQL, TypeScript, JavaScript
Machine Learning	PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, Optuna, SHAP
Deep Learning & GenAI	CNNs, ResNet50, YOLOv7, VGG19, UNet, OpenCV, LangGraph, RAG
NLP & AI Systems	Transformers (HuggingFace), spaCy, NLTK, ChromaDB, Vosk, Piper TTS
Backend & Data	FastAPI, REST APIs, WebSockets, PostgreSQL, SQLite, MySQL, Firebase Auth
Tools & DevOps	Git, GitHub Actions, Docker, Vercel, Render, Electron, PyInstaller

RESEARCH EXPERIENCE

Research Intern

Jun 2024 – Jul 2024

Thapar Institute of Engineering and Technology

- Benchmarked custom DCNN, VGG19-UNet, and ResNet50-UNet for binary crack classification on a 40,000-image dataset (**ResNet50: 99.9% validation accuracy**); augmentation pipeline used CLAHE, RandomResizedCrop, Otsu thresholding, and ColorJitter.
- Integrated the winning model into a real-time **YOLOv7** detection pipeline (NVIDIA RTX 2080 Ti, **97.7% mAP** over 100+ epochs) with automated, IRC:SP:83-aligned crack-width alerting via OpenCV; co-authored the resulting paper, submitted to **Elsevier** and currently under review.

PROJECTS

Ashwatthama — Local-First AI Desktop Assistant *Electron · React · FastAPI · Vosk · Piper TTS · ChromaDB*

- Architected a fully offline, three-layer system (Electron/React frontend, FastAPI backend, local inference layer) over WebSockets and a JWT-authenticated IPC bridge, with sub-200ms streaming ASR (**Vosk**) and <300ms voice-cloned TTS (**Piper TTS**).
- Built a **ChromaDB** vector store for semantic conversation memory (top-K cosine retrieval) and hardened every session with Fernet (AES-128-CBC + HMAC-SHA256) encryption and per-session JWT rotation.

TalentRank — Explainable Resume Screening *FastAPI · React · TypeScript · Sentence-Transformers · Firebase*

- Rebuilt an academic NLP prototype into a deployed product: a five-dimension explainable scorer (skills, experience, education, relevance, location) that returns evidence for every score, with live weight-tunable ranking and a bias-reduced blind-review mode.
- Shipped semantic matching (sentence-transformers with a TF-IDF fallback), a Firebase-authenticated **FastAPI** backend, and a Kanban hiring pipeline with full audit logging.

ArthSetu — Government Business-Identity Platform

FastAPI · React · TypeScript · SQLAlchemy

- Built with a 2-person team as **Product & Governance Workflow Engineer**, owning the human-in-the-loop review queue and governance workflows for a government-scale identity-resolution system.
- Designed explainable UBID matching (confidence-scored, evidence-backed auto-links) and an activity-inference engine classifying businesses as Active/Dormant/Closed from real operational signals.

ACHIEVEMENTS

- Top 6%** in Zelestra × AWS ML Ascend Challenge 2025; **Top 25** global rank in Shell.ai Hackathon 2025.
- Completed **Amazon ML Summer School 2026** (certified); solved **200+** **LeetCode** problems across graphs, trees, dynamic programming, and core data structures.
- Motivated by reducing catastrophic risks from advanced AI strong interest in LLM behavior evaluation, mechanistic interpretability, adversarial robustness, and AI alignment.